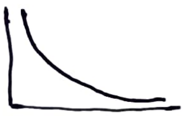
NEET - 2026 (Sample Paper - 03)

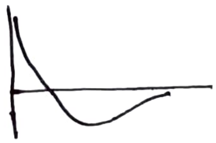
Number of Questions : 45 (Chemistry), Full Marks = 180
Marks for each correct response : 4 and incorrect response : -1

Q.1. 50 ml 0.5 M HCl solution is completely neutralised by 1.32 g of a sample of $CaCO_3$. The percentage purity of the sample is

1. 100   2. 95   3. 90   4. 75

Q.2. Which of the followings graphs correctly represent the variation of 'ψ' with 'r' for 2s orbital?

1. 

2. 

3: 

4. 

Q.3. Match the list-I with List-II

List-I		List-II
A. Element with highest electronegativity	(I)	I_2
B. Element with highest electron affinity	(II)	Br_2
C. Liquid non-metal	(III)	Cl_2
D. Solid non-metal	(IV)	F_2

Choose the correct answers from the options given below:

1. A-I, B-II, C-III, D-IV   2. A-III, B-IV, C-II, D-I

3. A-IV, B-III, C-I, D-II   4. A-IV, B-III, C-II, D-I

Q.4. Increasing order of bond length of O_2, O_2^-, O_2^{2-} and O_2^+ is

1. $O_2^+ < O_2 < O_2^- < O_2^{2-}$ 2. $O_2^{2-} < O_2^- < O_2 < O_2^+$

3. $O_2 < O_2^+ < O_2^- < O_2^{2-}$ 4. $O_2^+ < O_2^- < O_2^{2-} < O_2$

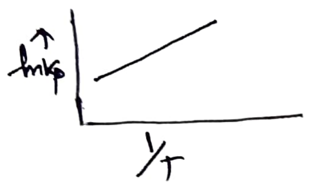
Q.5. 2 mole of an ideal gas at 27°C temperature is expanded reversibly from 2L to 20L. The entropy change in Cal. is ($R = 2$ Cal $K^{-1} mol^{-1}$)-

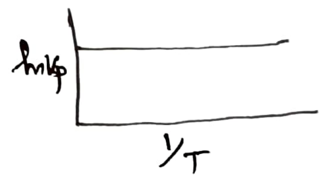
1. 92.1   2. 9.2   3. 0   4. 0.92

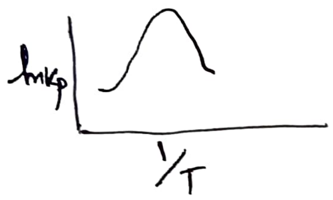
Q.6. If 'α' is the degree of dissociation of K_2SO_4, the van't Hoff factor (i) used to calculate the molar mass is

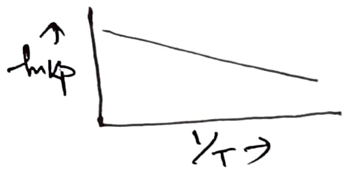
1. $1+\alpha$ 2. $1-\alpha$ 3. $1+2\alpha$ 4. $1-2\alpha$

Q.7. Which of the following graphs represents an exothermic reaction?

1. 

2. 

3. 

4. 

Q.8. Consider the change in oxidation state of bromine corresponding to different E° values as shown below:

$$BrO_4^- \xrightarrow{1.82\,V} BrO_3^- \xrightarrow{1.5\,V} HBrO \xrightarrow{1.595\,V} Br_2 \xrightarrow{1.0652\,V} Br^-$$

The species that undergoes disproportionation is

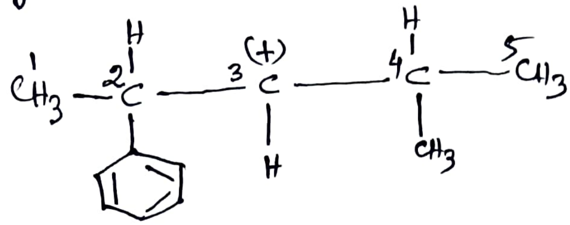
1. BrO_3^- 2. BrO_4^- 3. HBrO   4. Br_2

Q.9. The stability of +1 oxidation state increases in the sequence

1. Ga < In < Al < Tl   2. Al < Ga < In < Tl

3. Tl < In < Ga < Al   4. In < Tl < Ga < Al.

Q.10. In the following carbocation, H/CH_3 that is most likely to ~~be~~ migrate to the positively charged carbon is -



1. CH_3 at C-4
2. H at C-2
3. CH_3 at C-2
4. H at C-4

Q.11. Given below are two statements :

Statement I : But-1-yne forms white precipitate with ammoniacal silver nitrate solution while But-2-yne does not.

Statement II : But-1-yne has acidic hydrogen.

In the light of the above statements choose the most appropriate answer from the options given below :

1. Both Statement-I and Statement-II are correct.
2. Both Statement-I and Statement-II are incorrect.
3. Statement-I is correct but Statement-II is incorrect.
4. Statement-I is incorrect but Statement-II is correct.

Q.12. Given standard electrode potentials -

$$Fe^{2+} + 2\bar{e} \rightarrow Fe, \quad E^0 = -0.44\ V$$

$$Fe^{3+} + 3\bar{e} \rightarrow Fe, \quad E^0 = -0.036\ V$$

The standard electrode potential (E^0) for -

$$Fe^{3+} + \bar{e} \longrightarrow Fe^{2+} \text{ is}$$

1. -0.476 V
2. -0.404 V
3. $+0.404$ V
4. $+0.772$ V

Q.13. The rate of a gaseous reaction is halved when volume of the vessel is doubled. The order of the reaction is

1. 0 2. 1 3. 2 4. 3

Q.14. The reason why Conc. H_2SO_4 is used largely to prepare other acids is that Conc. H_2SO_4

1. Is highly ionised
2. Is strong oxidising agent
3. Is dehydrating agent
4. has high boiling point.

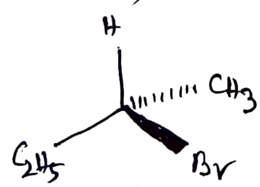
Q.15. $Cr_2O_7^{2-} \rightleftharpoons 2CrO_4^{2-}$ is shifted to right in

1. Acidic medium
2. Basic medium
3. Neutral medium
4. any medium.

Q.16. Which of the following is an outer orbital complex?

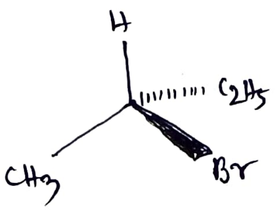
1. $[Fe(CN)_6]^{3-}$ 2. $[Fe(CN)_6]^{4-}$ 3. $[FeF_6]^{3-}$ 4. $[Cr(CN)_6]^{3-}$

Q.17. Which of the following structures is enantiomeric with the molecule (A) given below:

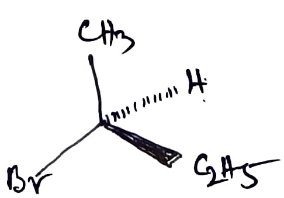


(A)

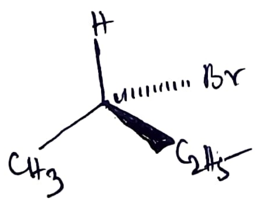
1.



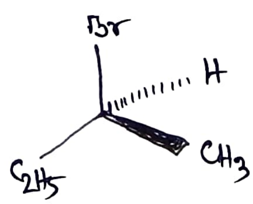
2.



3.



4.



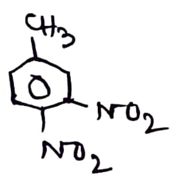
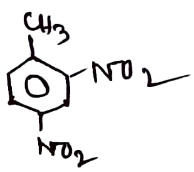
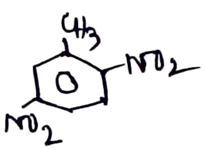
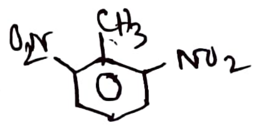
Q.18. 2-Phenylethanol may be Prepared by the reaction of Phenyl magnesium bromide with–

1. HCHO
2. 
3. CH_3COCH_3
4. CH_3CHO

Q.19. The appropriate reagent for the transformation

1. Zn(Hg), HCl
2. NH_2NH_2, OH^-
3. H_2/Ni
4. $NaBH_4$

Q.20. p-nitrotoluene on further nitration gives

1. 
2. 
3. 
4. 

Q.21. Number of chiral Carbon atoms in β-D(+) glucose is

1. 3
2. 4
3. 5
4. 6

Q.22. Lassaigne's test for the detection of nitrogen fails in

1. $NH_2CONH \cdot NH_2 \cdot HCl$
2. $NH_2NH_2 \cdot HCl$
3. NH_2CONH_2
4. $C_6H_5NHNH_2 \cdot HCl$

Q.23. When One mole of acidified $K_2Cr_2O_7$ reacts with excess KI, then number of ~~molecules~~ moles of I_2 liberated is

(a) 1. 2 2. 3 3. 6 4. 7

Q.24. In which orbital diagram both the Pauli's exclusion Principle and Hund's rule are violated?

1. [↑↓] [↑↑|↑|] 2. [↑↓] [↑↓|↑↓|↑↓]

3. [↑↓] [↓|↓|] 4. [↑↓] [↑↓|↑↓|↑]

Q.25. The formation of $O^{2-}(g)$ from $O(g)$ is 603 $KJ\ mol^{-1}$. If electron affinity of $O(g)$ is $-141\ J\ mol^{-1}$, the second electron affinity of oxygen would be

1. $+744\ KJ\ mol^{-1}$ 2. $-744\ KJ\ mol^{-1}$

3. $+462\ KJ\ mol^{-1}$ 4. $-462\ KJ\ mol^{-1}$.

Q.26. Which of the following is a Correct set with respect to molecule, hybridization and shape?

1. $BeCl_2$, sp^2, linear
2. $BeCl_2$, sp^2, trigonal Planar
3. BCl_3, sp^2, trigonal Planar
4. BCl_3, sp^3, tetrahedral.

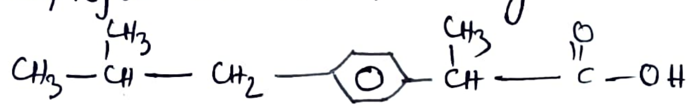
Q.27. A Current of dry air was bubbled through a bulb Containing 26 g of an organic substance in 200 g of water, then through a bulb Containing pure water at the same temperature and finally through a bulb Containing fused $CaCl_2$. The loss of weight of water bulb is 0.013 g and the gain of weight of of $CaCl_2$ tube is 4 g. The molecular weight of the organic substance is

1. 820 2. 740 3. 720 4. 840

Q.28. In which of the following reactions, the equilibrium remains unaffected on addition of small amount of argon at constant volume ?

1. $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
2. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
3. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
4. Equilibrium will remain unaffected in all the three cases.

Q.29. Ibuprofen has the following structure -



The compound shows

1. Geometrical isomerism
2. Optical isomerism
3. Tautomerism
4. Both (1) and (2)

Q.30. Among the following, the anti-aromatic compound(s) is/are :

(i) (ii) (iii) (iv)

1. (i), (ii), (iii)
2. (ii), (iii), (iv)
3. (i), (iv)
4. (i), (iii), (iv)

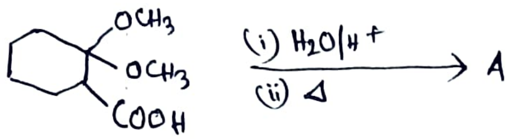
Q.31. For a certain reaction, the activation energy is zero. What is the value of rate constant at 300 K, if $K = 1.6 \times 10^6\ s^{-1}$ at 280 K?

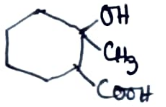
1. $1.6 \times 10^6\ s^{-1}$
2. Zero
3. 1
4. $3.2 \times 10^{12}\ s^{-1}$

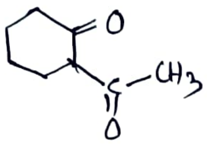
Q.32. The coordination number and the oxidation state of the element E in the complex $[E(en)_2(C_2O_4)]NO_2$ are, respectively

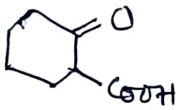
1. 6 and 2
2. 4 and 2
3. 4 and 3
4. 6 and 3

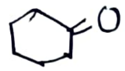
Q.33. Identify 'A' in the following reaction:



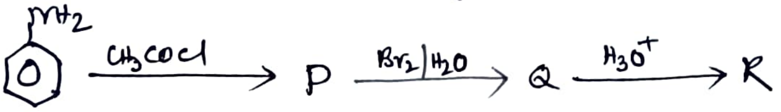
(1) 

(2) 

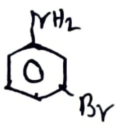
(3) 

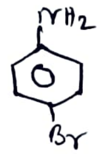
(4) 

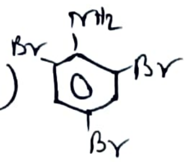
Q.34. Identify 'R' in the following reaction.



(1) 

(2) 

(3) 

(4) 

Q.35. What is the structure of Alanine at pH value 2?

(1) $\overset{+}{N}H_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COO^-$

(2) $NH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COO^-$

(3) $\overset{+}{N}H_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$

(4) Can be both (1) and (2).

Q.36. Match the List-I with List-II.

List-I		List-II
A.	Fe^{2+}	I. Blue ppt. with $K_3[Fe(CN)_6]$
B.	Fe^{3+}	II. Brown ppt. with $K_4[Fe(CN)_6]$
C.	Zn^{2+}	III. Bluish white ppt. with $K_4[Fe(CN)_6]$
D.	Cu^{2+}	IV. Blue ppt. with $K_4[Fe(CN)_6]$

Choose the correct answer from the options given below:

1. A-I, B-II, C-III, D-IV
2. A-I, B-IV, C-III, D-II
3. A-IV, B-III, C-I, C-II
4. A-III, B-IV, C-II, D-I

37. The nodal plane in the π-bond of ethene is located in

1. The molecular plane
2. A plane parallel to the molecular plane
3. A plane perpendicular to the molecular plane which bisects the carbon-carbon σ-bond at right angle.
4. A plane perpendicular to the molecular plane which contains the carbon-carbon bond.

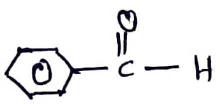
Q.38. Density of 2.05 M solution of acetic acid in water is 1.02 g/ml. The molality of the solution is

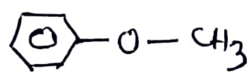
1. 1.14 2. 3.28 3. 2.28 4. 0.44

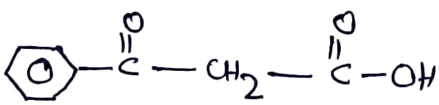
Q.39. 0.005 M acid solution has pH = 5. The percentage ionization of the acid is:

1. 0.8% 2. 0.6% 3. 0.4% 4. 0.2%

Q.40. Keto-enol tautomerism is observed in

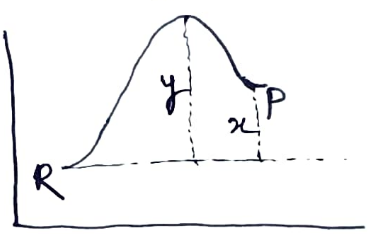
1. 

2. 

3. 

4. 

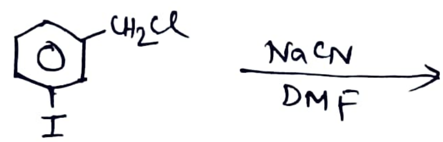
Q.41. The potential energy diagram for a reaction R→P is given below:

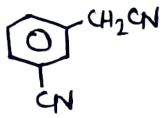


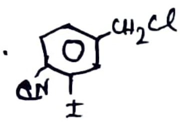
ΔH^0 of the reaction corresponds to the energy

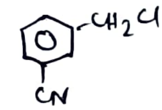
1. x 2. y 3. $(y-x)$ 4. $(x+y)$

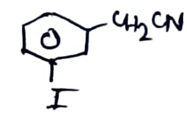
Q.42. The Structure of the major Product formed in the given reaction is :

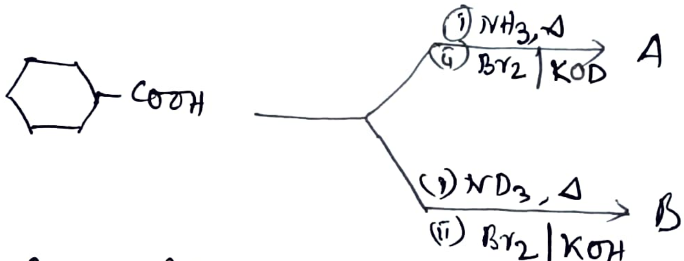


1. 

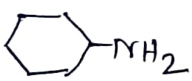
2. 

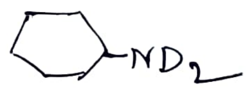
3. 

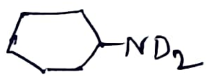
4. 

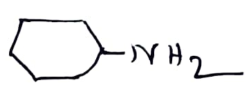
Q.43. 

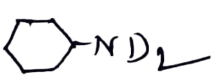
The Compounds A and B respectively are -

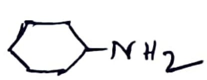
1. 

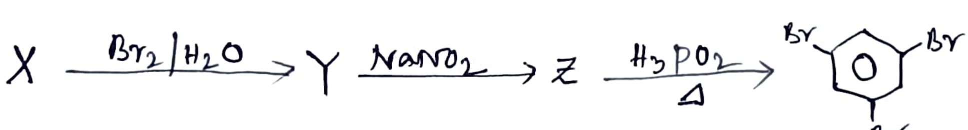
and 

2. 

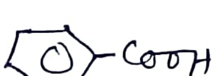
and 

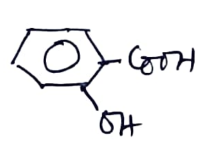
3. Both are 

4. Both are 

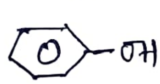
Q.44. 

In the above reaction, Compound 'X' is -

1. 

2. 

3. 

4. 

Q.45. The blue Colour obtained in the Lassaigne test is due to the formation of the Compound.

1. $Fe_4[Fe(CN)_6]$
2. $Fe_4[Fe(CN)_6]_3$
3. $Fe_3[Fe(CN)_6]_4$
4. $Na_4[Fe(CN)_6]$.

ANSWER KEYS : (NEET-2026, S.P-01)

Qn. NO	Correct option	Qn. NO	Correct option
1	2	26	3
2	2	27	3
3	4	28	4
4	1	29	2
5	2	30	2
6	3	31	1
7	1	32	4
8	3	33	4
9	2	34	3
10	2	35	3
11	1	36	2
12	4	37	1
13	2	38	3
14	4	39	4
15	2	40	3
16	3	41	1
17	1	42	4
18	2	43	2
19	2	44	3
20	2	45	2
21	3		
22	2		
23	2		
24	1		
25	1		